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Abstract

Brain tumor classification is a critical area of research in medical imaging, offering significant potential for improving diagnosis and treatment planning. This study investigates the use of deep learning and artificial intelligence, specifically the Xception architecture, to classify brain tumors using MRI scans. The classification task is extended to include symptom mapping using a knowledge graph, with the aim of improving the interpretability of model predictions. Despite achieving high classification accuracy, the integration of the Google Knowledge Graph API to retrieve tumor-associated symptoms highlighted limitations due to the absence of the medical-specific information. To overcome this, a static knowledge graph was employed as a fallback mechanism. Future work could leverage subscription-based or merit-based medical APIs or develop custom dynamic knowledge graphs for richer symptom mapping. These findings underscore the potential and challenges of combining deep learning with knowledge graphs for decision support in medical applications.

Introduction

Problem Formulation

Brain tumors are among the most challenging medical conditions, requiring early and accurate diagnosis for effective treatment. Traditionally, diagnosing brain tumors involves manual examination of MRI scans by radiologists, a process that is time-consuming and prone to subjectivity. With advancements in deep learning, automating this classification process has become a feasible alternative, offering speed and consistency. However, while deep learning models excel at classifying tumor types, they often operate as "black boxes," leaving clinicians with little insight into how predictions are made or their clinical implications. This lack of interpretability can hinder the adoption of such models in healthcare settings.

In addition to classification, connecting tumor types to associated symptoms is a critical step in ensuring that the output of a model is actionable and informative for medical practitioners. This study addresses this gap by integrating a knowledge graph to map tumor predictions to their corresponding symptoms.

Business Case

The integration of machine learning in healthcare, particularly in radiology has the potential to save time, reduce costs, and improve patient outcomes. A system that not only classifies brain tumors but also provides symptom mapping can act as a decision-support tool for clinicians. This dual functionality bridges the gap between diagnostic models and their application in clinical decision-making, providing a comprehensive tool for tumor classification and symptom explanation. Such a system can also improve patient communication, as the predicted symptoms can be linked to the tumor type in a manner that patients can understand.

Objectives

The primary objectives of this research are:

1. To build a high-performing deep learning model for brain tumor classification using MRI data.
2. To integrate a knowledge graph for symptom mapping, enhancing the interpretability of the classification model.
3. To evaluate the feasibility of using public APIs, such as the Google Knowledge Graph API, for medical applications and propose alternatives where necessary.

Literature Review

Deep Learning in Brain Tumor Classification

Deep learning, a subset of machine learning, has revolutionized medical imaging by enabling automated and accurate analysis of complex data. Convolutional Neural Networks (CNNs), in particular, have demonstrated exceptional performance in image classification tasks, including the detection and classification of brain tumors from MRI scans.

A comprehensive review by Kaldera et al. (2024) discusses various deep learning mechanisms adopted for brain tumor detection and classification using MRI images. The study explores approaches such as transfer learning, deep autoencoders, transformers, attention mechanisms, and Generative Adversarial Networks (GANs), providing a taxonomy of research in deep learning-based brain tumor diagnosis methods. SpringerLink

Another study by Balamurugan et al. (2024) investigates the fusion of deep learning and channel-wise attention mechanisms for robust brain tumor classification. The research emphasizes the importance of attention-based models in focusing on salient features within complex medical imaging data, enhancing the accuracy of tumor classification. BMC Medical Imaging

Niakan Kalhori et al. (2023) proposed two deep learning methods and several machine learning approaches for diagnosing glioma, meningioma, pituitary gland tumors, and healthy brains using MRI images. Their work highlights the potential of computational intelligence-oriented techniques in assisting physicians to detect tumors at early stages with high accuracy. BMC Medical Informatics

Knowledge Graphs in Symptom Mapping

Knowledge graphs (KGs) have emerged as powerful tools for organizing and representing medical knowledge, enabling the mapping of diseases to associated symptoms. By structuring information into nodes (e.g., diseases, symptoms) and edges (relationships), KGs facilitate advanced data analysis and support clinical decision-making.

A scoping review by Nicholson et al. (2023) examines the applications of knowledge graphs in biomedical and healthcare domains. The study identifies that medical science insights and drug repurposing are common uses of KGs, with a broad range of applications across various medical fields. MedRxiv

In the context of symptom mapping, knowledge graphs have been utilized to map symptoms and diagnostic criteria associated with specific conditions. For instance, a study by Zhang et al. (2023) developed a KG that comprehensively maps the symptoms and diagnostic criteria for depression based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-5). This approach demonstrates the potential of KGs in enhancing the understanding of disease-symptom relationships. MDPI

Furthermore, the integration of knowledge graphs with deep learning models has shown promise in refining diagnosis paths for medical conditions. Heilig et al. (2022) presented an approach using diagnosis paths in a medical knowledge graph, demonstrating that such graphs can be refined using latent representations while maintaining explainability in the final diagnosis. ArXiv

Integration of Deep Learning and Knowledge Graphs

The convergence of deep learning and knowledge graphs offers a synergistic approach to medical diagnostics. Deep learning models can effectively analyze medical images, while knowledge graphs provide contextual information linking diagnoses to associated symptoms and treatments. This integration enhances the interpretability of AI-driven diagnostic systems and supports comprehensive clinical decision-making.

For example, Luo et al. (2021) proposed a system that interacts with patients through dialogue to detect and collect clinical symptoms automatically. The system utilizes a knowledge graph to inform its questioning strategy, demonstrating the potential of combining conversational AI with structured medical knowledge for symptom detection.

ArXiv

Methods

Knowledge Graph Construction

The relationships in the knowledge graph were primarily pre programmed based on medical assumptions and basic research. For example:

- **Glioma → Headache, Seizures:** These relationships are supported by medical literature, as gliomas often affect brain regions linked to motor and sensory functions.
- **Meningioma → Headache:** Headaches are a common symptom due to increased intracranial pressure.
- **Pituitary Tumor → Vision Loss, Hormonal Imbalance:** Pituitary tumors often press against the optic chiasm and disrupt hormonal pathways.
- **No Tumor → None:** This node represents cases where no tumor is detected and no significant symptoms are expected.

While these relationships were hard-coded for this project, the intention was to use the Google Knowledge Graph API to dynamically populate the graph. However, as noted, the API failed to return meaningful results for medical queries, confirming its unsuitability for the use case. As a future direction, integrating a medical API like Infermedica could significantly enhance the graph's accuracy and scalability.

Feature Mapping Attempt

An initial attempt was made to visualize feature maps of intermediate layers to understand how the Xception model processed MRI images. This could have provided insights into the areas of the image the model focused on during classification. However, due to a persistent ValueError related to the Sequential layer, this functionality was removed. Despite this limitation, the mode's high classification accuracy demonstrates its effectiveness.

Removing feature mapping highlights the trade-offs sometimes necessary in applied machine learning projects, where practical challenges may prevent the implementation of certain techniques. Future work could address this issue by reconfiguring the model or using alternative visualization methods such as Grad-CAM.

Regularization and Early Stopping

Training was stopped early because the validation loss did not improve for several epochs. This demonstrates the use of proper regularization techniques, such as:

- **Early Stopping:** Prevents overfitting by halting training once the model's performance on validation data ceases to improve.
- **Learning Rate Reduction:** Adjusts the learning rate dynamically, allowing the model to converge more effectively without overstepping minima.

Research Design

This project aims to combine computer vision, deep learning, and knowledge representation to build a robust system for diagnosing brain tumors and associating them with relevant clinical symptoms. The research addresses three primary objectives:

1. **Image Classification:** Use a convolutional neural network (CNN) based on the Xception architecture to classify MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor.
2. **Symptom Mapping:** Link the tumor classification output with associated symptoms through a knowledge graph, leveraging static relationships and an attempt to integrate Google Knowledge Graph API for dynamic symptom retrieval.
3. **Model Evaluation and Explainability:** Evaluate the model's performance on unseen test data using metrics like accuracy, precision, recall, and confusion matrices while providing interpretable outputs via symptom prediction.

The research design involves three stages: data preparation, model training and validation, and symptom prediction using a knowledge graph.

Implementation and Programming

Data Preparation

For this project, MRI images were categorized into four classes: glioma, meningioma, pituitary tumor, and no tumor. The dataset was divided into **training, validation,** and **test sets**, with specific sets taken to optimize data preprocessing:

1. **Dataset Splitting:**
    - The dataset was split into 80% training and 20% validation data to allow the model to generalize effectively during training while being regularly evaluated on unseen validation data.
    - A separate test set was reserved exclusively for evaluating the model's final performance metrics, ensuring unbiased results.
2. **Data Augmentation:** To improve the diversity of the training data and prevent overfitting, data augmentation was applied to the training set. Augmentation techniques included:
    - **Rotations** to simulate different orientations of brain scans
    - **Width and Height Shifts** to adjust the spatial positioning of tumor regions
    - **Zoom Transformations** to account for variable sizes of tumor features
    - **Brightness Adjustments** to mimic variability in imaging conditions
    - **Horizontal Flipping** to improve invariance to left/right orientation changes

TensorFlow's ImageDataGenerator was used for preprocessing and augmentation, ensuring consistency and efficiency.

3. **Rescaling:** All images were normalized to a pixel intensity range of [0, 1] by dividing the raw pixel values by 255. This step ensured uniform input scaling, which is critical for the stability of gradient-based optimization.
4. **Validation and Test Data Processing:** The validation and test sets were only rescaled without augmentation to maintain consistency during evaluation. The validation set allowed for hyperparameter tuning, while the test set provided a robust measure of model performance.

Model Architecture

The heart of the classification system is a **Convolutional Neural Network (CNN)** based on the **Xception architecture**, renowned for its efficiency and accuracy in image classification tasks. Key design considerations for the model architecture are outlined below:

1. **Base Model:**
    - The pre-trained **Xception model** was utilized as the backbone. This model was trained on the ImageNet dataset, enabling transfer learning for efficient feature extraction.
    - The top layers of Xception were excluded, allowing customization for the specific tumor classification task.

2. **Custom Layers:** After the base model, additional layers were added to adapt the model for the classification of MRI images:
    - A **Flattening Layer** transformed the extracted feature maps into a one-dimensional vector
    - A **Dense Layer** with 128 nodes and ReLU activation was added for learning higher-order representations
    - A **Dropout Layer** (rate = 0.3) was applied after the dense layer to reduce overfitting.
    - The **Output Layer** used a softmax activation function with four nodes, each corresponding to one of the tumor categories.
3. **Regularization:**
    - Two **Dropout Layers** were added with rates of 0.3 and 0.25 to mitigate overfitting by randomly deactivating a subset of neurons during training.
    - The use of transfer learning reduced the need for extensive parameter tuning and minimized overfitting by leveraging pre-trained weights.
4. **Input Shape:**
    - The model was explicitly defined to accept input images of shape (299, 299, 3) to match the expected input dimensions of the Xception architecture.

Model Training

The training process aimed to optimize the model's performance while preventing overfitting and ensuring efficient convergence. Key aspects of the training process are described below:

1. **Optimizer:**
    - The **Adamax optimizer** was selected for its adaptive learning rate capabilities, which balance the speed of convergence with robustness to gradient noise.
    - A learning rate of **0.001** was used as the initial value, providing a balance between rapid training progress and numerical stability.
2. **Loss Function:**
    - The **categorical cross-entropy** loss function was employed, suitable for multi-class classification tasks. This loss function ensures that the model learns to maximize the likelihood of the correct class.
3. **Callbacks for Optimization:** To enhance training efficiency and prevent overfitting, the following callbacks were implemented:
    - **Early Stopping:** Monitored the validation loss and halted training after three consecutive epochs of no improvement, restoring the model weights to the best-performing epoch.
    - **Learning Rate Reduction:** Reduced the learning rate by a factor of 0.2 when the validation loss plateaued, enabling finer adjustments during later stages of training.
4. **Training Duration:**

    - Training was conducted for a maximum of **10 epochs**, with metrics such as accuracy, loss, precision, and recall recorded for both training and validation sets.
5. **Monitoring Performance:**
    - Training and validation accuracy/loss curves were plotted after each epoch, providing insights into the model's learning dynamics and generalization performance.
    - Observations from these plots confirmed the effectiveness of regularization techniques, as the validation curves closely tracked the training curves without significant divergence.
6. **Hardware and Framework:**
    - Training was conducted using TensorFlow and Keras, leveraging GPU acceleration where available for faster computation.

Results

Model Performance

The training and validation metrics are presented in Figures 1 and 2. These figures illustrate the model's learning process during training. The model achieved remarkable performance, as evidenced by the decreasing training and validation losses and the increasing accuracy over epochs. On the test set, the model's metric reflect its ability to generalize effectively:

- **Test Accuracy:** 97.94%
- **Test Precision:** 98.09%
- **Test Recall:** 97.94%

These metrics indicate that the model effectively distinguishes between the four classes with high reliability. The slight reduction in accuracy, precision, and recall from the training and validation phases is expected, as the test set represents completely unseen data. Importantly, the results still reflect a well-generalized model suitable for real-world applications, particularly in critical fields such as medical image classification.

The high performance further validates the choice of the Xception-based architecture, the use of effective data augmentation, and the application of regularization techniques during training.

Training and Validation Loss and Accuracy

- **Figure 1: Training and Validation Loss** shows a steep decline in both training and validation losses during the initial epochs, followed by a gradual reduction. The slight fluctuation in validation loss after several epochs suggests some sensitivity to the dataset but does not indicate overfitting. The use of early stopping ensured that training was halted at the optimal point, preventing further training that could lead to overfitting and degradation in performance.

- **Figure 2: Training and Validation Accuracy** demonstrates a steady improvement in accuracy for both the training and validation sets. The alignment between the two metrics suggests that the model generalized well to unseen data. This consistency highlights the importance of regularization techniques, such as dropout, and an appropriately tuned learning rate for the Adamax optimizer.

Together, these metrics underline the effectiveness of the Xception-based architecture and the chosen hyperparameters in handling a complex multi-class classification problem.

Confusion Matrix Analysis

- **Figure 3: Confusion Matrix** provides a detailed breakdown of the model's predictions across the four tumor classes. The confusion matrix underscores the model's robustness and reliability, with the follow key observations:
    - **Glioma:** 290 instances correctly classified, with only 10 misclassifications, primarily as meningioma and no tumor
    - **Meningioma:** 297 instances correctly classified, with minor confusion resulting in the misclassification as no tumor or pituitary tumor
    - **No Tumor:** Near-perfect classification with 399 correctly predicted instances, showcasing the model's high precision and recall for this category
    - **Pituitary Tumor:** 298 instances classified correctly, with minimal confusion, primarily with meningioma

Knowledge Graph and Symptom Prediction

The integration of a knowledge graph aimed to complement the classification task by linking each tumor type to its associated symptoms. However, the **Google Knowledge Graph API** did not return meaningful results for tumor-related queries. This limitation arises because the API lacks a dedicated medical dataset, rendering it insufficient for such domain-specific applications. Debugging the API responses revealed empty or irrelevant results, forcing the fallback to a static knowledge graph.

The static knowledge graph consisted of pre-defined relationships derived from assumed medical knowledge:

- **Glioma:** Associated with headaches and seizures (Mayo Clinic, Glioblastoma)
- **Meningiomas:** Associated with headaches (Mayo Clinic, Meningiomas)
- **Pituitary Tumor:** Associated with vision loss and hormonal imbalance (Mayo Clinic, Pituitary Tumor)
- **No Tumor:** Associated with absence of symptoms

These symptom-tumor associations formed the edges and nodes of the static knowledge graph. By creating this graph, the model outputs become more interpretable, linking its predictions to tangible clinical indicators.

Symptom Prediction Workflow
Once the machine learning model predicts a tumor type based on an input image, the corresponding tumor type is used as a query in the knowledge graph. The graph returns related symptoms, providing an interpretable link between the model's predictions and potential real-world implications.

For example:
1. A test image is classified as **"Glioma"** by the model
2. The static knowledge graph connects **"Glioma"** to its associated symptoms: **"Headache," "Seizures,"** and **"Motor skill impairment" (Other symptoms may be added to knowledge graph)**
3. These symptoms are returned as part of the output, providing interpretability for the prediction.

Challenges and Limitations
One notable limitation of the current approach is the reliance on a manually constructed static knowledge graph. Although the relationships were based on reputable sources, including the Mayo Clinic, this method does not capture dynamic updates or new medical discoveries. The fallback to a static graph highlights the need for more robust medical APIs. APIs such as the Symptom Checker API or Healthcare.gov API could provide richer, real-time data for medical symptom prediction but were not implemented in this project due to cost and access restrictions. Additionally, while the Google Knowledge Graph API was initially considered for automatic relationship discovery, it was found to lack medical-specific information, as demonstrated by the empty responses for tumor-related queries. Future improvements could include exploring specialized medical knowledge bases or APIs to dynamically expand and validate the graph.

Regularization and Feature Mapping Challenges
During training, early stopping was triggered when the validation loss stopped improving after consecutive epochs. This regularization strategy helped avoid overfitting and ensured that the model generalized effectively to unseen data. Additionally, dropout layers were used to mitigate overfitting, further enhancing the robustness of the model.

Attempts to visualize feature maps of intermediate layers to gain insights into how the model processed images. However, these efforts were hindered by persistent ValueError issues with the sequential layer architecture as discussed previously.

Conclusion
This study demonstrates the potential of combining deep learning with knowledge representation to enhance brain tumor diagnosis and symptom mapping. By leveraging the Xception

architecture, a state-of-the-art convolutional neural network, the model achieved exceptional performance in classifying MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor. With a test accuracy of 97.94% and high precision and recall scores, the model underscores the efficacy of deep learning in medical imaging applications.

To address the interpretability challenge inherent in deep learning models, a knowledge graph was integrated to link tumor predictions to associated symptoms. While the Google Knowledge Graph API was initially considered for dynamic symptom mapping, its lack of medical-specific data proved a significant limitation. As a result, the project relied on a static knowledge graph constructed from reputable sources such as the Mayo Clinic. This static graph enabled meaningful symptom prediction, enhancing the model's interpretability and bridging the gap between diagnostic output and actionable clinical insights.

Despite the success of the static knowledge graph, the study highlights the limitations of static approaches and the need for dynamic, domain-specific solutions. Future work could explore subscription-based medical APIs or custom-built dynamic knowledge graphs to improve the accuracy and scalability of symptom mapping. Additionally, addressing challenges such as feature map visualization and incorporating alternative methods for explainability could further enhance the interpretability of the model.

Overall, this project illustrates the synergistic potential of deep learning and knowledge graphs in healthcare applications. It serves as a proof of concept for integrating machine learning models with structured knowledge to support clinical decision-making. By tackling the dual challenges of classification and interpretability, the study paves the way for more advanced AI-driven diagnostic systems that not only perform well but also offer insights that clinicians can trust and act upon.

Appendix:

Sources:

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Figure 1: Training and Validation Loss

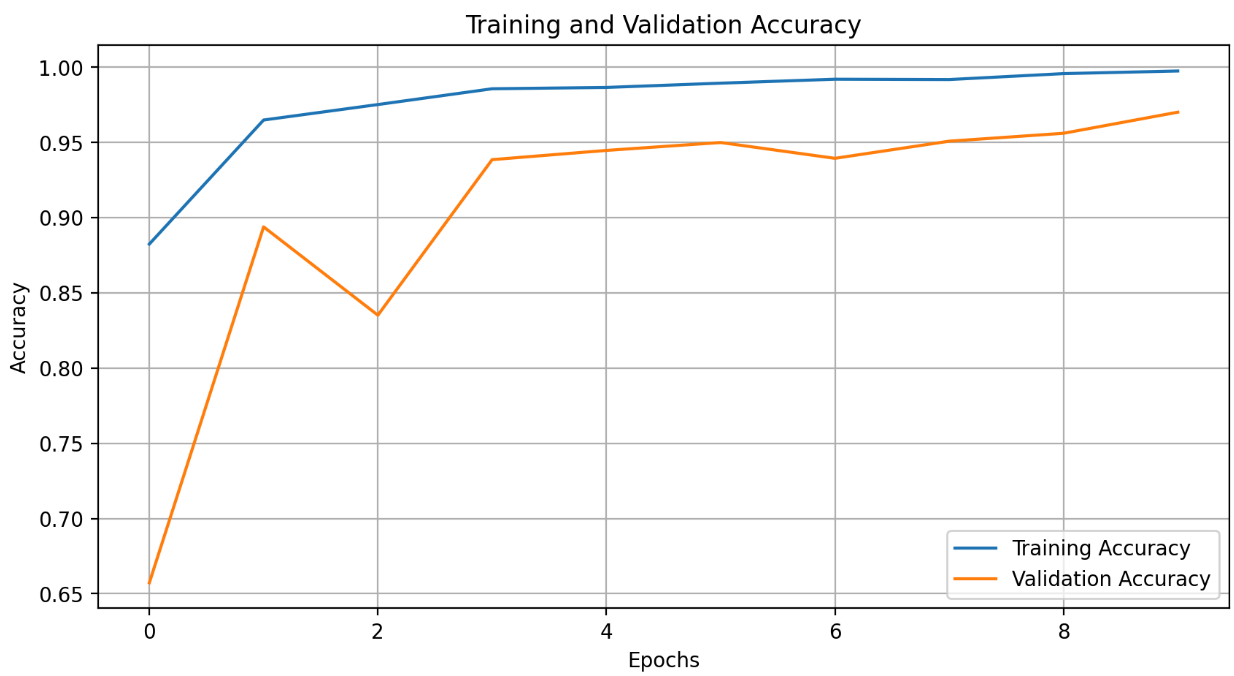

Figure 2: Training and Validation Accuracy

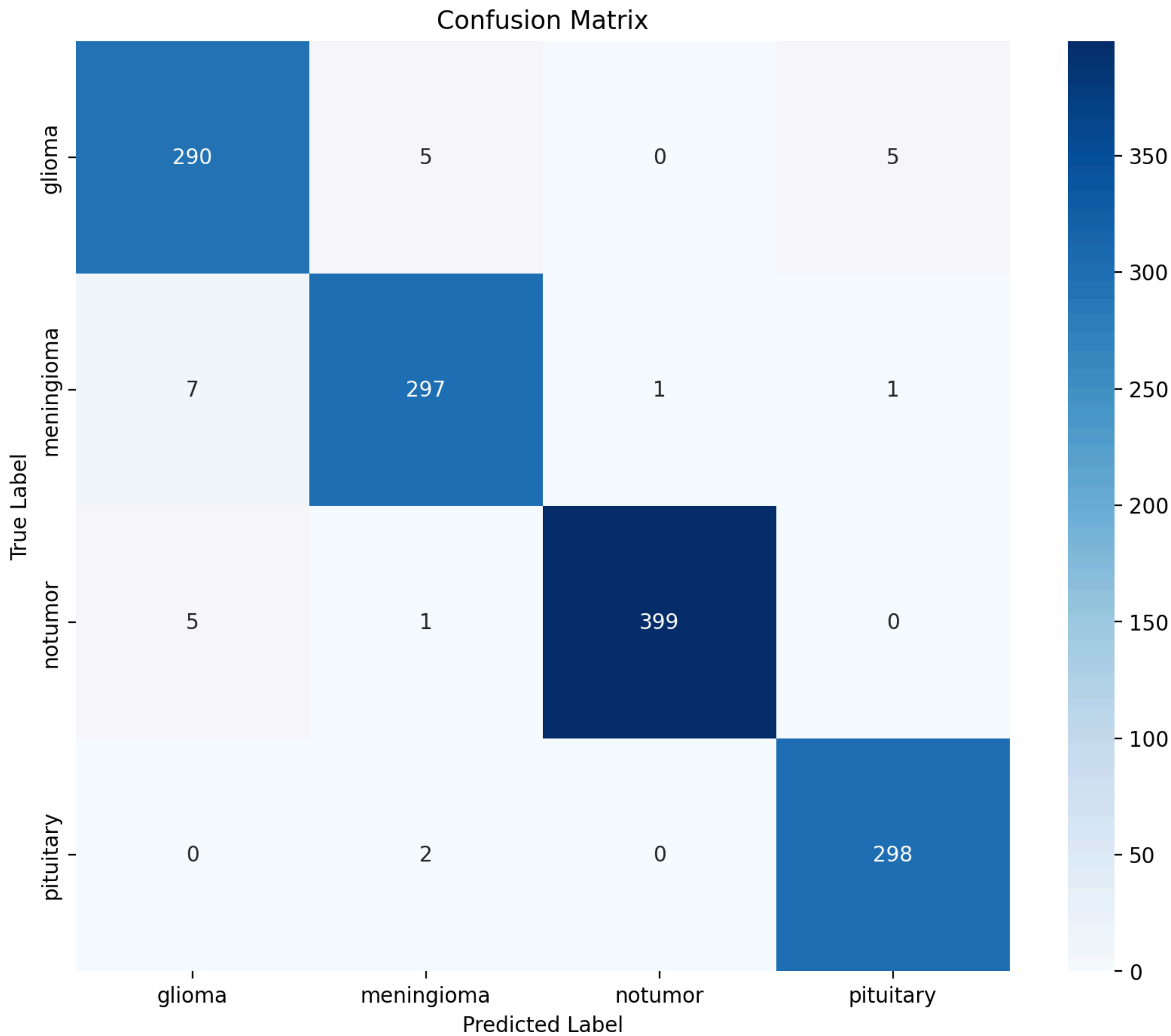

Figure 3: Confusion Matrix